

Saddle-point problems and the inf-sup condition

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REFERENCES

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Benzi–Golub–Liesen, *Acta Numerica*, 2005.

Saddle-point problems in Hilbert spaces

Consider the following problem in saddle-point form¹

$$\begin{cases} a(u, v) - b(v, p) = g(v) & \forall v \in V, \\ -b(u, q) = -f(q) & \forall q \in Q, \end{cases}$$

with

- ▶ V, Q Hilbert spaces wrt the products $(\cdot, \cdot)_V, (\cdot, \cdot)_Q$;
- ▶ $a : V \times V \rightarrow \mathbb{R}$ bilinear, continuous, **symmetric**;
- ▶ $b : V \times Q \rightarrow \mathbb{R}$ bilinear and continuous.

The problem can be viewed as a **constrained minimization problem**: find

$$u = \operatorname{argmin}_{v \in V} \left[\frac{1}{2} a(v, v) - g(v) \right]$$

subject to the linear constraint

$$b(v, q) = f(q) \quad \forall q \in Q.$$

¹This includes mixed Poisson / Darcy as a model example.

Lagrangian and saddle point

Introduce the Lagrangian

$$\mathcal{L}(v, q) := \frac{1}{2}a(v, v) - g(v) - b(v, q) + f(q).$$

Then the mixed problem is equivalent to finding a saddle point:

$$\inf_{q \in Q} \sup_{v \in V} \mathcal{L}(v, q) = \mathcal{L}(u, p) = \sup_{v \in V} \inf_{q \in Q} \mathcal{L}(v, q).$$

Theorem (Equivalence between saddle-point and constrained minimiser)

If a is symmetric positive definite, then (u, p) solves the symmetric mixed problem if and only if (u, p) is a saddle point of $\mathcal{L}$.

First-order optimality conditions

Let us verify the Euler–Lagrange conditions. Consider variations around (u, p) :

$$\mathcal{L}(u+tv, p+sq) = \frac{1}{2}a(u+tv, u+tv) - g(u+tv) - b(u+tv, p+sq) + f(p+sq).$$

Differentiating at $(t, s) = (0, 0)$ gives

$$\left. \frac{d}{dt} \mathcal{L}(u + tv, p) \right|_{t=0} = a(u, v) - g(v) - b(v, p),$$

$$\left. \frac{d}{ds} \mathcal{L}(u, p + sq) \right|_{s=0} = -b(u, q) + f(q).$$

Hence the stationary conditions are exactly

$$a(u, v) - b(v, p) = g(v) \quad \forall v \in V, \quad -b(u, q) = -f(q) \quad \forall q \in Q.$$

Linear algebra viewpoint

In matrix form, the discrete system has the block structure

$$\mathbb{A} = \begin{pmatrix} A & B^T \\ B & 0 \end{pmatrix},$$

with $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{m \times n}$.

Theorem (symmetric and semidefinite case)

If A is symmetric positive semidefinite and B has full rank m , then

- ▶ $\mathbb{A}$ is nonsingular if and only if

$$\ker(A) \cap \ker(B) = \{0\},$$

- ▶ $\mathbb{A}$ is indefinite, with positive and negative eigenvalues.

Corollary (SPD case)

If A is SPD and $\text{rank}(B) = m$, then $\mathbb{A}$ is invertible.

Why the inf-sup condition appears

For A and B stemming from a mixed PDE discretization of the saddle-point problem, uniqueness of the discrete linear system is not sufficient; one needs a mesh-uniform discrete inf-sup condition to ensure stability and convergence. **Stability in the subspace $\Rightarrow$ error bounds.**

The matrix may still be invertible on each mesh, but the method is not robust:

- ▶ pressure may become unstable,
- ▶ spurious pressure modes may appear,
- ▶ velocity may lock,
- ▶ error constants blow up (convergence is not guaranteed in a uniform way.)

If inf-sup does not hold, one may need to stabilise the system in the form of a 2-2 block, leading to stabilised system matrix

$$\mathbb{A} = \begin{pmatrix} A & B^T \\ B & C \end{pmatrix},$$

Towards the inf-sup condition

Define the **kernel**² of b :

$$Z := \{v \in V : b(v, q) = 0 \ \forall q \in Q\}.$$

Then the mixed problem is equivalent to:

- first solve: find $u \in Z$:

$$a(u, v) = g(v) \quad \forall v \in Z,$$

- then recover p from: find $p \in Q$:

$$b(v, p) = a(u, v) - g(v) \quad \forall v \in V.$$

The first step is well posed if a is coercive on Z . **The second step is the real issue**: it requires compatibility between V and Q through b ...

²In the incompressible fluids case, the space of divergence-free vectors.

Operator interpretation

Define the **weak residual** r from

$$b(v, p) = \underbrace{a(u, v) - g(v)}_{=:(r, v)_V} \quad \forall v \in V.$$

As $(r, z)_V = 0$, $\forall z \in Z$, we have

$$r \in Z^\perp := \{v \in V : (v, z)_V = 0, \forall z \in Z\}$$

Introduce the operator

$$B : V \rightarrow Q', \quad \langle Bv, q \rangle := b(v, q).$$

Then the pressure-recovery equation is

$$Bv = r \in Z^\perp.$$

So solvability amounts to requiring that B has sufficiently large range.

Example: Darcy case

In the Darcy case,

$$B = -\operatorname{div} : H(\operatorname{div}; \Omega) \rightarrow L^2(\Omega), \quad B^t = \nabla : L^2(\Omega) \rightarrow H^{-1}(\Omega)^d.$$

The pressure problem reads: find $p \in Q = L^2(\Omega)$:

$$\nabla p = r \quad \text{with } r \in Z^\perp.$$

Hence, this is **solvable** iff $\operatorname{range}(\nabla Q) = Z^\perp$, which is clearly true in the continuous case...

Brezzi's theorem

Theorem (Saddle-point problem wellposedness)

Let V, Q be Hilbert spaces, and let $a : V \times V \rightarrow \mathbb{R}$, $b : V \times Q \rightarrow \mathbb{R}$ be continuous bilinear forms. Define

$$Z := \{v \in V : b(v, q) = 0 \ \forall q \in Q\}.$$

If

- ▶ a is coercive on Z ,
- ▶ there exists $\beta > 0$ such that

$$\sup_{v \in V \setminus \{0\}} \frac{b(v, q)}{\|v\|_V} \geq \beta \|q\|_Q \quad \forall q \in Q,$$

then the saddle-point problem is well posed.

The second condition is the [Babuška–Brezzi–Ladyzhenskaya inf-sup condition](#). It is required to hold in the discrete setting to ensure optimal convergence...

Continuous Darcy revisited

The continuous mixed Darcy problem fits Brezzi's theorem with

$$V = H(\operatorname{div}; \Omega), \quad Q = L^2(\Omega).$$

The inf-sup condition can be justified using the Poisson problem: for any $q \in L^2(\Omega)$, solve

$$-\Delta \phi = q, \quad \phi \in H_0^1(\Omega),$$

and set

$$u = \nabla \phi.$$

Then

$$\nabla \cdot u = q, \quad \|u\|_{H(\operatorname{div}; \Omega)} \leq C \|q\|_{L^2(\Omega)}.$$

So B is surjective and has a continuous lifting.

Moreover, $\forall v \in Z$,

$$a(v, v) = \int_{\Omega} K^{-1} |v|^2 \geq \alpha \|v\|_{[L^2(\Omega)]^d}^2 = \alpha \|v\|_{H(\operatorname{div}; \Omega)}^2$$

that is, a is coercive on Z .

Discrete inf-sup is not automatic

The continuous inf-sup condition does not imply the discrete one.
It depends on the compatibility between the discrete spaces $V_h \subset V$ and $Q_h \subset Q$.

For Raviart–Thomas, the crucial properties are:

$$\nabla \cdot RT_k = Q_h,$$

and the restriction to RT_k of the divergence operator

$$B_h := B|_{RT_k} : RT_k \rightarrow Q_h$$

is surjective. In particular,

$$\ker(B_h) \subset \ker(B).$$

The RT interpolant also satisfies a commuting property:

$$\nabla \cdot (I_h^{RT} v) = \Pi_h(\nabla \cdot v) \quad \Rightarrow \quad \|\nabla \cdot (I_h^{RT} v)\|_{L^2(\Omega)} = \|\Pi_h(\nabla \cdot v)\|_{L^2(\Omega)} \leq \|\nabla \cdot v\|_{L^2(\Omega)}$$

where Π_h is the L^2 projection onto Q_h .

Fortin's trick (A standard tool for proving discrete inf-sup.)

Lemma (Fortin's trick)

Let $V_h \subset V$, $Q_h \subset Q$. Assume the continuous inf-sup condition holds:

$$\sup_{v \in V \setminus \{0\}} \frac{b(v, q)}{\|v\|_V} \geq \beta \|q\|_Q \quad \forall q \in Q.$$

If there exists an operator $\Pi_h^F : V \rightarrow V_h$ with $\|\Pi_h^F v\|_V \leq C_\Pi \|v\|_V$, with $C_\Pi > 0$ constant independent of h , and

$$b(v - \Pi_h^F v, q_h) = 0 \quad \forall v \in V, \forall q_h \in Q_h,$$

then for $\tilde{\beta} = \frac{\beta}{C_\Pi}$, a constant clearly independent of h , it holds

$$\sup_{v_h \in V_h \setminus \{0\}} \frac{b(v_h, q_h)}{\|v_h\|_V} \geq \tilde{\beta} \|q_h\|_Q \quad \forall q_h \in Q_h.$$

Proof.

$$\sup_{v_h \in V_h} \frac{b(v_h, q_h)}{\|v_h\|_V} \geq \sup_{v \in V} \frac{b(\Pi_h^F v, q_h)}{\|\Pi_h^F v\|_V} = \sup_{v \in V} \frac{b(v, q_h)}{\|\Pi_h^F v\|_V} \geq \sup_{v \in V} \frac{b(v, q_h)}{C_\Pi \|v\|_V} \geq \frac{\beta}{C_\Pi} \|q_h\|_Q$$

Why RT satisfies Fortin's criterion

The Raviart–Thomas interpolation operator *almost* does the whole job:

- ▶ For v sufficiently regular,

$$b(v - I_h^{RT} v, q_h) = 0 \quad \forall q_h \in Q_h,$$

thanks to the commuting property.

- ▶ It satisfies the stability estimate

$$\|I_h^{RT} v\|_{H(\operatorname{div}; \Omega)} \leq C \|v\|_{H^1(\Omega)},$$

in $H^1(\Omega)$ but *not* on $H(\operatorname{div}; \Omega)$...

Discrete inf-sup condition for RT: direct proof

Take $q_h \in Q_h$. By surjectivity of the continuous divergence operator, there exists $v \in [H^1(\Omega)]^d$ such that

$$\nabla \cdot v = q_h, \quad \|v\|_{H^1(\Omega)} \leq C \|q_h\|_{L^2(\Omega)}.$$

Now define

$$v_h := I_h^{RT} v \in RT_k.$$

Using the commuting property,

$$\nabla \cdot v_h = \nabla \cdot (I_h^{RT} v) = \Pi_h(\nabla \cdot v) = \Pi_h q_h = q_h.$$

Hence

$$b(v_h, q_h) = \int_{\Omega} (\nabla \cdot v_h) q_h \, dx = \|q_h\|_{L^2(\Omega)}^2.$$

Moreover,

$$\|v_h\|_{H(\operatorname{div}; \Omega)} \leq C \|v\|_{H^1(\Omega)} \leq C \|q_h\|_{L^2(\Omega)}.$$

Therefore

$$\sup_{w_h \in RT_k} \frac{b(w_h, q_h)}{\|w_h\|_{H(\operatorname{div}; \Omega)}} \geq c \|q_h\|_{L^2(\Omega)}.$$